

- Confirm H_0 : $\Box H_0$,
- Confirm H_1 : $\Box H_1$, and
- Neither Confirm H_0 or H_1 : $\Diamond H_0 \wedge \Diamond H_1$.

The truth condition for each of the possible outcomes of a test can be given directly in terms of the set-theoretic relationship between the (prespecified) confidence interval process c , evidence E , and hypotheses H_0 and H_1 . Intuitively, we say that a hypothesis is confirmed provided that a sufficiently leveled confidence interval is *fully contained within* H_i . The three possible set-theoretic relationships can be visualized as follows:

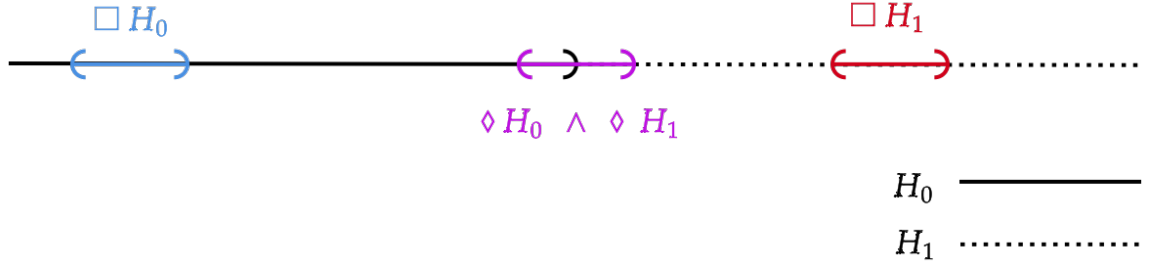


FIGURE 1. Satisfaction Conditions for Confirmation

More formally, we have

- (1) $\Box H_i$ if and only if $c(E) \subseteq H_i$, and
- (2) $\Diamond H_0 \wedge \Diamond H_1$ if and only if $c(E) \cap H_0 \neq \emptyset$ and $c(E) \cap H_1 \neq \emptyset$.

Confirmation of a hypothesis H_i in this sense is therefore witnessed by a sufficiently narrow, well-located confidence region. Indecisiveness of the test occurs when the confidence interval is not fully contained within *either* H_0 or H_1 .

Reducing the problem of hypothesis testing to containment properties of confidence regions results in a particularly natural and simple semantics for a multivalent logic of hypothesis testing. In the remainder of this section we illustrate how to think of the possibility of confirmation through the lens of hypothesis testing through three examples.

2.3.2. Two-Sided Hypotheses. To begin with an example, consider $\Theta = \mathbb{R}$ and suppose

$$(3) \quad H_0 : \theta = \theta_0 \text{ vs. } H_1 : \theta \neq \theta_0$$

We reject H_0 provided that the $1 - \alpha$ confidence interval excludes $\{\theta_0\}$:

$$(4) \quad \text{Reject } H_0 \Leftrightarrow c(E) \cap \{\theta_0\} = \emptyset.$$

Likewise, by swapping the role of the null and alternative hypotheses,

$$(5) \quad \text{Reject } H_1 \Leftrightarrow c(E) \cap \{\theta_0\}^c = \emptyset.$$

The hypothesis H_0 can be rejected: if, for example, $c(E) = (\theta_0 + \epsilon, \theta_0 + \eta)$ with $\epsilon, \eta > 0$ then $c(E) \cap \{\theta_0\} = \emptyset$ and we can reject H_0 .

By contrast, H_1 *cannot* be rejected. There are many proofs of this, but I focus here on a topological one: the confidence interval $c(E)$ is a nonempty open subset of $\mathbb{R}$, and $\{\theta_0\}$ is a single point. Thus $\{\theta_0\}^c$ is a dense subset of $\mathbb{R}$ and therefore intersects $c(E)$, so H_1 cannot be rejected. This entails that H_0 cannot in principle be *confirmed* in this testing paradigm.

The above line of reasoning can be made more simple by converting it to the notation of modal logic. We begin by observing that “failure to reject H_0 ” expresses a kind of *possibility* or *plausibility*³:

³The interpretation of confidence intervals modally, as a range of plausible values for the parameter θ , has been advocated by Smithson [12]

the evidence E generating the confidence interval $c(E)$, when subjected to the test T (i.e. $c(E) \subseteq H_1$), left open the possibility of H_0 (at least at the level α). Thus, we get the following set of equivalences.

$$\begin{aligned} \Diamond H_0 &\Leftrightarrow \neg(\text{Reject } H_0) \\ &\Leftrightarrow \neg(c(E) \cap H_0 = \emptyset) \\ &\Leftrightarrow (c(E) \cap H_0 \neq \emptyset) \\ &\Leftrightarrow \neg(c(E) \subseteq H_1) \end{aligned}$$

Adopting the typical modal definition $\Box = \neg\Diamond\neg$ we get a definition for the *confirmation* of the null hypothesis given evidence:

$$\begin{aligned} \Box H_0 &\Leftrightarrow \neg\Diamond\neg H_0 \\ &\Leftrightarrow \neg\Diamond H_1 \\ &\Leftrightarrow \neg(\neg(c(E) \subseteq H_0)) \\ &\Leftrightarrow c(E) \subseteq H_0. \end{aligned}$$

In plainer language, one confirms the null H_0 just in case one rejects the alternative H_1 . This means that confirmation is a distinguished form of null result: one in which not only is the p -value of the test greater than α since $c(E) \cap H_0 \neq \emptyset$, but one in which the alternative hypothesis is simultaneously *rejected* at level α . Thus, confirmation is a more stringent condition than merely finding a null result.

However, the very possibility of the confirmation of a hypothesis hinges on the topological structure of the hypothesis. Returning to the original two-sided example, if $H_0 = \{\theta\}$ and $H_1 = \mathbb{R} \setminus H_0$ then

$$\models \neg\Box H_0;$$

i.e. it is a validity that H_0 is not confirmable relative to H_1 . Equivalently,

$$\models \Diamond H_1.$$

Two-sided hypotheses can therefore be considered non-confirmable in a precise sense.

2.3.3. Two One-Sided Hypotheses. Let $M > 0$. Consider the Two One-Sided Test (TOST) hypothesis

$$(6) \quad H_0 : |\theta - \theta_0| > M \text{ vs. } H_1 : |\theta - \theta_0| \leq M$$

We reject H_0 provided that the $1 - \alpha$ confidence interval excludes $(-\infty, -M) \cup (M, \infty)$:

$$(7) \quad \neg\Diamond H_0 \Leftrightarrow c(E) \cap H_0 = \emptyset.$$

By contrast with two-sided hypotheses, the TOST paradigm admits a nontrivial notion of confirmation of the null *and* the alternative:

$$(8) \quad \models \Box H_0 \Leftrightarrow c(E) \cap [-M, M] = \emptyset.$$

and

$$(9) \quad \models \Box H_1 \Leftrightarrow c(E) \cap ((-\infty, -M) \cup (M, \infty)) = \emptyset.$$

both of which are possible.

In the following section we build up a general framework for reasoning about this kind of confirmation.

3. FORMAL SEMANTICS FOR THE MODAL LOGIC OF HYPOTHESIS TESTING

3.1. Syntax. The salient features of this logic require only a very sparse syntax. We may suppose that we have a single sentence $\{H_0\}$, the negation connective $\{\neg\}$, and the possibility operator $\Diamond$. We define $H_1 = \neg H_0$ and $\Box H_i = \neg\Diamond\neg H_i$. We adopt the usual definitions of satisfaction for the remaining Boolean connectives, although they are not our primary focus.

3.2. Semantics. Let Θ be a space of parameters, and $H_0 \subset \Theta$. Let Ω be the sample space of observations. By evidence E , I mean a finite string of observations $E \in \Omega^{<\omega}$. The semantics of modal logics are commonly exhibited in terms of Kripke frames by defining an accessibility relation. Kripke semantics for the logic described in this section are provided in Appendix A. To keep the core notions of statistical confirmation accessible to a wide audience, we present a more direct equivalent semantics.

A *possible world* is a pair (θ, E) where $\theta \in \Theta$ and $E \in \Omega^{<\omega}$. Let $H_0 \subseteq \Theta$. We write

$$(10) \quad w = (\theta_w, E_w) \Vdash H_0 \Leftrightarrow \theta_w \in H_0.$$

Satisfaction of the negation of a sentence S is defined via:

$$(11) \quad w \Vdash \neg S \Leftrightarrow \neg(w \Vdash S).$$

To define the semantics on modal operators, we need a notion analogous to that of a confidence interval. By a confidence function I mean a function $c : \Omega^{<\omega} \rightarrow 2^\Theta$, which assigns to evidence E a subset $c(E) \subset \Theta$. At the outset we do not require anything further of c , not even that c be nonempty: we can interpret $c(E) = \emptyset$ as saying that the evidence E is incompatible with all $\theta \in \Theta$. However, we say that c is non-refuting provided $c(E) \neq \emptyset$ for all $E \in \Omega^{<\omega}$.

Given a confidence function c on $\Omega^{<\omega}$, we can define the semantics of possibility operators:

$$(12) \quad w \Vdash \Diamond H_0 \Leftrightarrow c(E_w) \cap H_0 \neq \emptyset.$$

Routine calculation shows that

$$(13) \quad w \Vdash \Box H_0 \Leftrightarrow c(E_w) \subseteq H_0.$$

Note that $\Box$ is generally irreflexive (i.e. the reflexive axiom $\Box\phi \rightarrow \phi$), underlying a key feature of statistical confirmation. In general,

$$(14) \quad w \Vdash \Box H \not\Rightarrow w \Vdash H.$$

For example, let $H_0 = \{\theta_0\}^c$ and $w = \{\theta_0, E\}$ such that $\theta_0 \notin c(E)$. Then $w \not\Vdash H$ since $\theta_w = \theta_0$, but $w \Vdash \Box H_0$ since $c(E) \subseteq H_0$ as $\theta_0 \notin c(E)$.

A final observation is that there are precisely three satisfiable outcomes for the satisfaction of the modal sentences:

Proposition 3.1. Let c be non-refuting. Then for every H_0 and world w , exactly one of the three sentences is true: $w \Vdash \Box H_0$, $w \Vdash \Box H_1$, or $w \Vdash \Diamond H_0 \wedge \Diamond H_1$.

Proof. It suffices to show that $\Box H_0 \wedge \Box H_1$ is false, since $\Diamond H_0 \wedge \Diamond H_1 \Leftrightarrow \neg(\Box H_0 \vee \Box H_1)$ and $P \vee Q \vee \neg(P \vee Q)$ is a propositional tautology.

Suppose for contradiction that $w \Vdash \Box H_0 \wedge \Box H_1$. Then $c(E_w) \subset H_0$ and $c(E_w) \subset H_0^c$, so $c(E_w) = \emptyset$. This contradicts c being non-refuting. $\square$

3.3. The Topological Case. In this section we add additional constraints to Θ and c in line with the topological properties of the confidence interval in parametric statistics. The reader unfamiliar with topological terminology is advised to consult the classic textbook [11] for the definitions of relevant terminology.

Definition 3.2. Let Θ be a topological space and $\mathcal{O}(\Theta)$ the collection of open sets. A open-valued confidence function is a map $c : \Omega^{<\omega} \rightarrow \mathcal{O}(\Theta)$.

Open-valued confidence functions are natural objects. A confidence function being open-valued is just to say that if θ is plausible with respect to E (i.e. $\theta \in c(E)$), then sufficiently small perturbations